

ENVX2001 - Model selection

Learning Outcomes

In this lab, you will work towards achieving learning outcomes

Lab Objectives

In this lab, we will:

- []
- []



Tip

Please work on this exercise by creating your own R Markdown file.

Exercise 1: Model quality - multiple vs adjusted r^2

Data: *California_streamflow* spreadsheet

Import the “California_streamflow” sheet into R.

CODE

```
# Load library if needed
library(readxl)
# Load data
stream_data <- read_xlsx("data/california_streamflow.xlsx", "streamflow")
```

in this exercise we will use the same data as last week. To jog your memory, the dataset contains 43 years of annual precipitation measurements (in mm) taken at (originally) 6 sites in the Owens Valley in California. Through model selection via partial F-test we have the final model as below:

```
CODE
fit <- lm(runoff_volume ~ rock_creek + pine_creek, data = stream_data)
```

We will now add a totally useless variable to the dataset. This variable is a random number generated from a normal distribution with mean 3 and standard deviation 2. We use the `set.seed()` function to make sure that everybody gets the same random values.

```
CODE
set.seed(100) # to make sure everybody gets the same results

# this generates the random number into the dataset
stream_data$random_no <- rnorm(n = nrow(stream_data), mean = 3, sd = 2)
```

We will see the impact of including a totally useless variable, such as this random variable, has on measures of model quality, r^2 and adjusted r^2 values.

Task: create two regression models:

1. `runoff_volume ~ rock_creek + pine_creek`
2. `runoff_volume ~ rock_creek + pine_creek + random_no`

Question 2

Compare each in terms of their multiple r^2 and adjusted r^2 values. Which performance measure (multiple r^2 or adj r^2) would you use to identify which predictors to use in your model?

Exercise 2: Fish productivity

Fish communities were surveyed in lakes across a eutrophication gradient to investigate the relationship between productivity and fish diversity.

The datasheet has the following variables:

- `lake_id` Unique identifier for each lake
- `chl_a` Log-transformed Chlorophyll a
- `richness` Log-transformed species richness
- `evenness` Log-transformed Pielou's evenness
- `abundance` Log-transformed abundance per unit effort (NPUE)
- `biomass` Log-transformed biomass per unit effort (BPUE)
- `productivity` Log-transformed productivity proxy

CODE

```
#Load library if necessary
library(tidyverse)

#Read in data
fish_data <- read_csv("data/fish_communities.csv")
```

OUTPUT

```
Rows: 39 Columns: 7
— Column specification —————
Delimiter: ","
chr (1): lake_id
dbl (6): chl_a, richness, evenness, abundance, biomass, productivity

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

CODE

```
#Have a look at the structure of the data
str(fish_data)
```

OUTPUT

```
spc_tbl_ [39 × 7] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ lake_id      : chr [1:39] "FTH" "XLH" "HGH" "LH" ...
 $ chl_a       : num [1:39] 4.61 4.53 3.84 3.71 2.43 ...
 $ richness     : num [1:39] 2.64 2.89 1.99 2.94 2.37 ...
 $ evenness    : num [1:39] -0.356 -0.67 -0.423 -0.405 -0.589 ...
 $ abundance    : num [1:39] 2.24 3.34 1.24 2.67 1.12 ...
 $ biomass     : num [1:39] 4.96 6.03 4.67 6.52 5.32 ...
 $ productivity: num [1:39] 3.47 4.47 2.71 4.47 3.24 ...
 - attr(*, "spec")=
 .. cols(
 ..   lake_id = col_character(),
 ..   chl_a = col_double(),
 ..   richness = col_double(),
 ..   evenness = col_double(),
 ..   abundance = col_double(),
 ..   biomass = col_double(),
 ..   productivity = col_double()
 .. )
 - attr(*, "problems")=<externalptr>
```

Explore the data on your own before making any models. (*Hint: remember that we are only interested in the numeric variables for our linear models*)

Question 1

Are there any obvious relationships between the response variable (productivity) and the other variables?

Question 2

Are there any other patterns or potential issues you can see from the exploratory plots?

Take home exercise:

Review

Attribution